

Highlights

Geometric Observables for Financial Regime Detection

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- Walk-forward Berry Phase Rate achieves unbiased OOS Cohen's $d = 0.72$ with 30% fewer false alarms than Random Forest
- Four geometric observables from a single QCML embedding detect regime shifts without crisis labels
- Offline separability on 17 crises (2000–2024): Reduced Purity $d = 0.83$, Absorption Ratio $d = 0.80$ (classical benchmark), Berry Phase Rate $d = 0.61$ (rank 9 of 46)
- Formal framework with three theorems validated empirically on 20 years of market data

Geometric Observables for Financial Regime Detection

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Abstract

We embed financial time series into a projective Hilbert space via spectral metric learning (QCML) and extract four geometric observables—Berry Phase Rate, Spectral Entropy, Reduced State Purity, and Hamiltonian Sensitivity—that detect regime transitions without labeled data. We test them alongside classical baselines on 17 crises spanning 2000–2024.

Walk-forward nested HPO, our primary evaluation, gives Berry Phase Rate an unbiased out-of-sample Cohen’s $d = 0.72$ (5/9 test windows) with 30% fewer false alarms than Random Forest (2.5 vs. 3.6/yr). In the offline comparison, Reduced State Purity ($d = 0.83$) ranks first of 46 methods, with Absorption Ratio [1] ($d = 0.80$) as the classical benchmark; the remaining geometric channels provide mid-range separability ($d = 0.53$ – 0.61) but contribute orthogonal crisis coverage. Three theorems characterizing the embedding geometry are validated empirically.

A companion paper extends this framework to the full nineteen-channel observatory with orthogonality analysis and lead-time experiments; a third paper develops adaptive fusion strategies.

Keywords: spectral metric learning, regime detection, Berry curvature, quantum Fisher information, Fubini–Study metric, geometric observables, financial crises, QCML

1. Introduction

Financial markets switch between calm trends and violent drawdowns. Detecting these regime transitions is a core problem in quantitative finance. Classical approaches—Hidden Markov Models [2, 3], Bayesian Online Changepoint Detection [4], supervised classifiers—work well in specific settings but operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [5] embeds data as unit vectors in $\mathbb{CP}^{n-1}$ and represents features as Hermitian operators. This endows the data manifold with a Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [5] proved that the metric’s eigenvalue spectrum captures intrinsic manifold dimension; Abanov et al. [6] developed the mathematical theory. The geometric observables we extract measure deformations of a manifold whose structure $g(x)$ faithfully captures data geometry (Section 2.1).

Classical computation. Everything runs on classical hardware. The “quantum” label refers to the mathematical formalism (Hilbert space, Hermitian operators), not to quantum computing.

We extract four geometric observables from the QCML embedding—Berry Phase Rate, Spectral Entropy, Reduced State Purity, and Hamiltonian Sensitivity—and compare them against classical baselines including Absorption Ratio, Hamilton Markov-Switching, and GARCH (Table 2).

Our primary evaluation is *crisis-window separability*: does the score distribution during a known crisis separate from normal? This is an offline event study—it measures sensitivity, not real-time prediction. Walk-forward evaluation (Section 5.2) tests causal detection with strictly past-fitted preprocessing.

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Supervision level. Score construction is unsupervised—no crisis labels needed for z-scores (Algorithm 1). But the pipeline is *semi-supervised*: hyperparameters are selected via Optuna TPE with crisis labels as the optimization objective. Hilbert dimension ranges from $n = 4$ to $n = 16$ per observable; operator method is selected by ablation (Section 5.3).

Contributions.

1. **Walk-forward detection.** Nested HPO yields Berry Phase Rate OOS $d = 0.72$ (5/9 crises), with 30% fewer false alarms than RF (2.5 vs. 3.6/yr), without requiring crisis labels.
2. **QCML geometric framework.** Four geometric observables from a single embedding, each probing a different property of the data manifold (curvature, entropy, sensitivity, entanglement).
3. **Offline separability.** On 17 crises (2000–2024), Reduced Purity ($d = 0.83$) ranks first of 46 methods, with Absorption Ratio ($d = 0.80$) as the classical benchmark; the remaining geometric channels ($d = 0.53$ – 0.61) contribute orthogonal crisis coverage.
4. **Independent evaluation.** The first academic evaluation of QCML geometry outside the framework’s own developers [5, 6, 7, 8].

A caveat. The offline rankings measure sensitivity, not real-time detection. Reduced Purity ($d = 0.83$ offline) is sensitive to bipartition choice. Berry Phase Rate, which ranks lower offline ($d = 0.61$), gives the strongest walk-forward evidence. These channels augment, not replace, existing methods [9, 10].

1.1. Related Work

Several non-Euclidean frameworks have been applied to financial data (Table 1).

Information geometry and optimal transport. The Fisher–Rao metric [11] has been applied to interest rate manifolds [12] and return distributions [13]; Horvath et al. [14] cluster regimes via Wasserstein distance. Both provide metrics but lack Berry curvature, fiber bundle structure, and spectral gaps.

TDA and network curvature. Gidea and Katz [15] use persistence landscapes for crash detection (algebraic topology); Sandhu et al. [16] use Ollivier–Ricci curvature on correlation networks (the closest curvature-based comparator). Our Berry curvature operates on the data manifold rather than inter-asset graphs, yielding topological invariants via Chern–Weil theory.

Random matrix theory and signatures. Laloux et al. [17] separate signal from noise via Marchenko–Pastur; our spectral gap is a phase-transition order parameter, not a noise edge. Issa and Horvath [18] detect regimes via signature-based MMD tests; Bronstein et al. [19] impose geometric priors rather than discovering geometry from data.

Classical regime detection. Hamilton’s [2] Markov-switching, Ang and Bekaert’s [3] financial extensions, and Bollerslev’s [20] GARCH are all included as baselines.

QCML literature. The framework was developed by Candelori et al. [5] and Abanov et al. [6]; financial applications were proposed in industry papers [7, 8]. Our contribution is not the framework but its application: four geometric regime-detection channels, a head-to-head comparison against established baselines with proper statistical testing, and walk-forward out-of-sample validation.

2. QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator $A = A^\dagger$ [5, 6].

Given $x = (x_1, \dots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [5, 7] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^p (A_k - x_k I)^2, \quad (1)$$

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

Framework	Metric	Curvature	Gauge	Topol. Invt.	Spect. Gap
Info. Geometry	Fisher–Rao	Sectional	α -conn.	—	—
Optimal Transport	Wasserstein	—	—	—	—
TDA	—	—	—	Betti	—
Network Ricci	—	Ollivier	—	—	—
Random Matrix Thy.	—	—	—	—	MP edge
Rough Path Sig.	Sig. kernel	—	—	—	—
QCML (ours)	Fubini–Study	Berry	U(1)	Chern	$\Delta(x)$

where I is the $n \times n$ identity. The *quasi-coherent state* at x is the ground state:

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [21]:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where indices $a, b \in \{0, \dots, p-1\}$ run over PCA components, and the *Berry curvature* [22, 6]:

$$F_{ab}(x) = -2 \text{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (4)$$

Equation (4) holds in the parallel-transport gauge $\langle \psi | \partial_a \psi \rangle = 0$; our numerical implementation uses the gauge-invariant plaquette formula (Section 3.1).

Theorem 1 (Smoothness [5]). *If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ on an open set $U \subset \mathbb{R}^p$, then $x \mapsto |\psi(x)\rangle$, $g_{ab}(x)$, and $F_{ab}(x)$ are C^∞ on U .*

Proof sketch. By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle \psi | \psi \rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix Appendix A for the full argument. $\square$

Operator construction.. We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k} \sigma_k$. An alternative is *random* Hermitian operators: $A = (M + M^\dagger)/2$, where entries of M are drawn from a standard complex normal distribution. Random operators avoid the near-degeneracy that PCA-inspired operators can produce for certain Hilbert dimensions (see Section 5.3). All stochastic elements (random operator generation, bootstrap resampling, RF training) use seed 42 for reproducibility. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.3. Full QCML gradient-descent training [7, 8] is left for future work.

Why fixed operators suffice for detection.. The QCML framework [5] centers on learning operators via reconstruction loss to estimate intrinsic dimension and generate predictions. Our application—regime *detection*—has a weaker requirement: we need observables that respond to *changes* in the data geometry, not observables that optimally represent it. By Theorem 1, any operator set yielding a positive spectral gap $\Delta(x) > 0$ induces a C^∞ geometry whose metric, curvature, and topological invariants are well-defined. Distributional shifts (regime changes) perturb the Hamiltonian $H(x)$ and hence the induced geometry; the resulting observable responses are detectable regardless of whether the operators were optimized. The empirical ablation (Section 5.3) confirms this: heuristic operators (random and PCA-inspired) already achieve large effect sizes ($d > 0.8$) with no gradient training, and discriminative operator learning does not systematically improve detection performance.

Proposition 2 (Fisher information interpretation). *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor (3) satisfies $4g_{ab}(x) = [F_Q]_{ab}(x)$, where F_Q is the quantum Fisher information matrix [23, 24, 25]. Consequently:*

- (i) *The QFI pseudo-determinant $\det^+(g)$ is proportional to the volume element of the Fisher information metric on the statistical manifold: $\det^+(g) = 4^{-r} \det^+(F_Q|_r)$, where $r = \text{rank}(g)$.*
- (ii) *The quantum Cramér–Rao bound [26, 27] $\text{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements.*
- (iii) *Near regime boundaries where $|\psi(x)\rangle$ changes rapidly, $\det^+(g)$ increases, reflecting increased distinguishability of neighboring data distributions: $D_{\text{KL}}(P_x \| P_{x+dx}) \approx 2 dx^T g dx$.*

Parts (i)–(iii) follow from standard results in quantum estimation theory; see Appendix Appendix A.

2.1. Intrinsic Manifold Dimension

A natural question is *why* the quantum metric $g_{ab}(x)$ is the right geometric structure for financial regime detection. Candelori et al. [5] provide the answer: the eigenvalue spectrum of g_{ab} exhibits a spectral gap at position d equal to the intrinsic dimension of the data manifold. Crucially, this dimension estimate is robust to noise—unlike MLE, DANCo, and TwoNN estimators, which create “shadow dimensions” from noise artifacts, the QCML metric exploits spectral gap stability to filter noise eigenvalues. Our geometric observables therefore measure deformations of a manifold whose geometry $g(x)$ faithfully captures, connecting QCML to classical results in random matrix theory [17, 28] and factor models [29].

Empirical demonstration.. We compute the eigenvalue spectrum of $g_{ab}(x_t)$ on the SPY/DIA feature space at each time step and identify the spectral gap. In normal periods, the largest eigenvalue ratio $\lambda_2/\lambda_3 \approx 4.5$ indicates a sharp gap at $d \approx 2$ –3, consistent with Laloux et al. [17, 28] ($\sim 6\%$ signal eigenvalues) and Bai and Ng [29] (~ 2 latent factors). During crises, this ratio drops to ~ 2.8 and the metric participation ratio $\text{PR}(g) = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ rises from 2.63 ± 0.54 to 2.92 ± 0.57 (Cohen’s $d = 0.54$, $p < 10^{-56}$)—the manifold *unfolds* as herding breaks the normal low-dimensional structure (Figure 1). The QCML spectral gap dimension detects this shift with $d = 0.69$ ($p < 10^{-104}$), whereas PCA effective rank barely distinguishes regimes ($d = 0.13$)—a five-fold sensitivity advantage. Spectral Entropy (Section 3.2) tracks the full eigenvalue distribution and achieves the highest offline separability among geometric channels ($d = 0.83$).

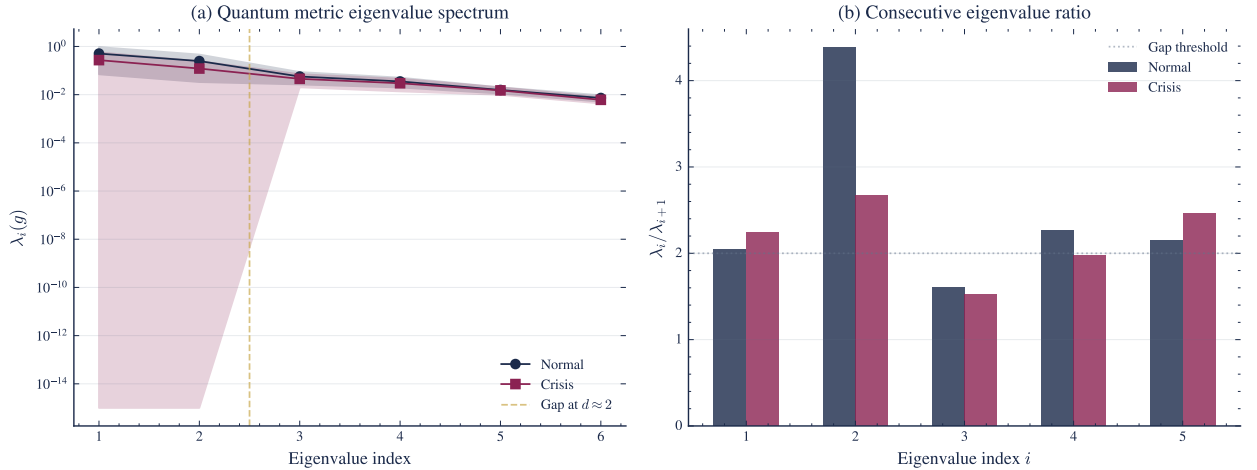


Figure 1: Quantum metric eigenvalue spectra at normal vs. crisis timepoints. (a) Mean spectrum with $\pm 1\sigma$ bands; the dashed line marks the spectral gap at $d \approx 2$. (b) Consecutive eigenvalue ratios λ_i/λ_{i+1} ; the sharp gap at $i = 2$ in normal periods (ratio ≈ 4.5) weakens during crises (ratio ≈ 2.8), indicating manifold dimension increase.

Connection to existing observables.. The Absorption Ratio [1]—which measures the fraction of variance explained by the top eigenvalue of the rolling correlation matrix—is a *linear* dimension measure (PCA eigenvalue concentration). Our quantum metric generalizes this nonlinearly: the QCML embedding maps data through $x \mapsto |\psi(x)\rangle$ into $\mathbb{CP}^{n-1}$, where the pullback metric captures curvature and topological structure absent in flat PCA space. Candelori et al. [5] show that gradient-descent-trained operators yield optimal manifold approximation; our pre-defined operator basis preserves the spectral gap structure relevant for regime detection, though absolute dimension estimates should be interpreted as approximate.

3. Geometric Observables

We focus on four geometric observables extracted from the QCML embedding, each computed at every time step t and converted to z-scores via Algorithm 1. Table 2 summarizes the channels; a fifth comparator—Absorption Ratio [1]—is a classical baseline included in Section 4 to calibrate the value added by geometric structure. A companion paper extends the observatory to nineteen channels spanning seven geometric families.

Table 2: Geometric observatory channels studied in this paper.

Channel	Family	Geometric property	Equation
Berry curvature rate	Holonomy	U(1) holonomy change rate	(5)
Spectral entropy	Spectral	Excitation distribution	(7)
Hamiltonian sensitivity	State	Perturbation variance	(8)
Reduced purity	State	Subsystem separability	(9)

3.1. Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (5)$$

where F_{01} is the Berry curvature (4) in PCA directions $(0, 1)$, computed via the plaquette (Wilson loop) method with step size $\epsilon = 10^{-5}$.

Hyperparameters.. Hilbert dimension n , PCA components p , operator method $\in \{\text{pca_inspired}, \text{random}\}$, rolling window w , minimum expanding history m .

Plaquette computation.. The Berry curvature F_{ab} is computed via the discrete plaquette (Wilson loop) method. For each pair of PCA directions (a, b) at point x :

$$F_{ab}(x) = -\frac{1}{\epsilon^2} \text{Im} \log(\langle \psi(x) | \psi(x + \epsilon e_a) \rangle \times \langle \psi(x + \epsilon e_a) | \psi(x + \epsilon e_a + \epsilon e_b) \rangle \langle \psi(x + \epsilon e_a + \epsilon e_b) | \psi(x + \epsilon e_b) \rangle \langle \psi(x + \epsilon e_b) | \psi(x) \rangle), \quad (6)$$

with $\epsilon = 10^{-5}$. This gauge-invariant formula avoids explicit phase-fixing [22, 30].

3.2. Spectral Entropy

Shannon entropy of excitation energy weights:

$$S(t) = -\sum_{n=1}^{N-1} w_n \log w_n, \quad w_n = \frac{E_n(t) - E_0(t)}{\sum_{m=1}^{N-1} (E_m(t) - E_0(t))}. \quad (7)$$

Low entropy indicates energy concentration in few excitations (ordered regime); high entropy signals a disordered/crisis regime. Among the four geometric channels studied here, Spectral Entropy achieves the highest offline separability ($d = 0.83$).

3.3. State Dynamics

Hamiltonian Sensitivity. Variance of the Hamiltonian perturbation:

$$\sigma_H^2(t) = \langle \psi(x_t) | \Delta H^2 | \psi(x_t) \rangle - \langle \psi(x_t) | \Delta H | \psi(x_t) \rangle^2, \quad (8)$$

where $\Delta H = H(x_t) - H(x_{t-1})$. Large sensitivity signals rapid structural change in the error Hamiltonian.

Reduced State Purity. Purity of the reduced density matrix:

$$P(t) = \text{tr}(\rho_A^2), \quad \rho_A = \text{tr}_B(|\psi(x_t)\rangle\langle\psi(x_t)|), \quad (9)$$

with bipartition A (dimension 2) and B (dimension $n/2$). Low purity signals loss of separability—the ground state cannot be decomposed into independent factor components, a signature of correlated crisis dynamics. Results depend on the bipartition choice; we use $(2, 4)$ at $n = 8$. The headline d value is conditional on this choice.

All four geometric observables share the same core hyperparameters: Hilbert dimension $n \in \{4, \dots, 16\}$, PCA components p , operator construction method, rolling window w , and minimum expanding history m . Values are tuned per observable via the HPO protocol of Section 4.2.

3.4. Z-Score Thresholding (Algorithm 1)

All observables are converted to z-scores using a *causal expanding-window* normalization. This is the “adaptive thresholding” referenced in Algorithm 3.

Algorithm 1 Causal Expanding-Window Z-Score

Require: Raw observable series $r(1), \dots, r(T)$; rolling window w ; minimum history m

Ensure: Z-score series $z(m), \dots, z(T)$

- 1: Smooth: $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$ for $t \geq w$
 - 2: **for** $t = m, \dots, T$ **do**
 - 3: $\mu(t) = \text{mean}(s(1), \dots, s(t-1))$ ▷ expanding mean of past only
 - 4: $\sigma(t) = \text{std}(s(1), \dots, s(t-1))$ ▷ expanding std of past only
 - 5: $z(t) = (s(t) - \mu(t)) / \sigma(t)$
 - 6: **end for**
-

Algorithm 2 FAR-Calibrated Threshold

Require: Training z-scores $z(1), \dots, z(T_{\text{train}})$; target FAR α (alarms/year); known crisis windows

Ensure: Threshold τ

- 1: Remove crisis-window indices from z to obtain z_{normal}
 - 2: Binary search: find smallest τ such that $\text{FAR}(z_{\text{normal}} > \tau) \leq \alpha$
 - 3: **return** τ
-

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.

Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

4. Experimental Protocol

4.1. Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (1998–2024) from Yahoo Finance via the `yfinance` library (v0.2.x). Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, min, max over a 20-day lookback), then reduced to $p = 15$ PCA components and L^2 -normalized.

We evaluate on 17 historical crises spanning 2000–2024, classified *a priori* as 9 Conventional (with historical parallels) and 8 Novel (unprecedented mechanisms). Crisis windows are extended by ± 10 trading days. Full definitions appear in Appendix G, Table G.10.

4.2. Methods and HPO Protocol

For the main comparison (Table 3), geometric methods use per-observable Hilbert dimensions ($n \in \{4, 6, 8, 16\}$) selected via dimension sensitivity analysis: observables whose d improves monotonically with dimension use $n = 16$, while others retain their HPO-optimized values found via Optuna TPE with consistency penalty (see below). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels. GARCH(1,1) [20] produces an expanding z-score of conditional variance; Hamilton Markov-switching [2] fits a 2-regime MS-AR(1) with switching variance and outputs $P(\text{high-variance regime})$. Turbulence Index [31] computes a multivariate Mahalanobis distance on the expanding feature matrix; Absorption Ratio [1] tracks the top-eigenvalue variance fraction of the rolling correlation matrix. The full hyperparameter search spaces for all methods appear in Appendix F, Table F.9.

Random Forest protocol. For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

Rolling RF (VIX) protocol. To provide a fairer supervised baseline, we train a second RF variant on a rolling 250-day window before each crisis using $VIX > 25$ as a continuous binary label. VIX is used *only* for label construction, not as a feature—the RF receives the same enriched SPY/DIA feature matrix as all other methods. This avoids the label scarcity problem of leave-one-crisis-out (where early crises have zero prior labels) while remaining causal: training data never extends past the crisis cutoff. We also include a VIX Level detector (expanding z-score of raw VIX close) as an oracle upper bound, since VIX directly measures implied volatility.

Per-crisis causal preprocessing. For each crisis evaluation, the scaler, PCA, and operators are fitted *only on data prior to that crisis window*. Specifically, the causal cutoff is set at crisis start minus 10 calendar days; all preprocessing (standardization, PCA projection, operator fitting) uses only rows $1, \dots, t_{\text{cutoff}}$. Scores are then computed on the full timeline. The 20-day enrichment lookback provides an additional implicit buffer of approximately 19 trading days, yielding an effective cutoff of ~ 33 trading days before crisis onset. Classical baselines are similarly trained only on pre-crisis data. RF uses leave-one-crisis-out labels restricted to the causal window. This eliminates lookahead in preprocessing.

Statistical evaluation. Each method–crisis pair: Cohen’s d with block-bootstrap 95% CI ($n = 10,000$; block size = $\lceil n^{1/3} \rceil$ per Politis & White [?]). Financial z-scores may violate the normality assumption underlying Cohen’s d ; the block bootstrap CIs and Cliff’s δ (computed alongside d as a distribution-free robustness check) provide nonparametric alternatives. Individual tests: Welch’s t -test (Holm–Bonferroni corrected), permutation test ($n = 5,000$). Multi-method: Friedman rank test with Nemenyi post-hoc pairwise comparisons at $\alpha = 0.05$.

HPO-optimized hyperparameters. The main results (Table 3) use hyperparameters selected via Optuna TPE (100 trials) with a consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$ on a pre-2020/post-2020 split. Selected configuration for Berry Phase Rate: $n=6$, $p=8$, $w=15$, random operators. Post-HPO dimension sensitivity analysis found that some observables benefit from higher n (up to 16); each retains its HPO-optimized or dimension-upgraded value. Full search spaces in Appendix F.

HPO optimism bias.. Nested cross-validation reveals a mean optimism bias of $+0.415d$; reported HPO-tuned d values should be interpreted as upper bounds. The walk-forward results (Section 5.2) provide causally valid estimates.

4.3. Detection Pipeline

Algorithm 3 QCML Geometric Regime Detection

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n , PCA components p

Ensure: Z-scored observable time series

- 1: Standardize X ; PCA to p components; L^2 -normalize
 - 2: Fit operators $\{A_k\}$ via PCA-inspired method (or expanding-window refit)
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Solve $|\psi(x_t)\rangle = \text{ground state of } H(x_t)$ (1)
 - 5: Compute raw observables (Table 2)
 - 6: **end for**
 - 7: Convert to z-scores via Algorithm 1
-

Computational cost.. Per-step eigensolve cost scales as $O(n^3)$. At $n = 16$, representative timings: Spectral Entropy $\sim 1\text{s}$; Hamiltonian Sensitivity $\sim 1\text{s}$; Reduced Purity $\sim 2\text{s}$ ($T = 3,447$, single CPU). Berry Phase Rate at $n = 6$: $\sim 0.5\text{s}$. Random Forest: 1.07s ($p = 15$).

5. Results

5.1. Crisis-Window Separability

Table 3 presents the aggregate comparison across all 17 crises using per-crisis causal preprocessing: for each crisis, scaler, PCA, and operators are fitted only on data *before* that crisis window (Section 4.2). HPO-optimized hyperparameters are used for all geometric methods; Random Forest uses leave-one-crisis-out cross-validation. Among the four featured geometric observables, Reduced State Purity achieves the highest median $d = 0.83$ (rank 1 of 46), followed by Berry Phase Rate (0.61, rank 9), Hamiltonian Sensitivity (0.60, rank 10), and Spectral Entropy (0.53, rank 12). Absorption Ratio (0.80, rank 2) provides the classical benchmark; Hamilton Markov-switching ranks 3rd (0.71), Random Forest 30th (0.35), and GARCH(1,1) 33rd (0.29). The Friedman test ($\chi^2 = 233.1$, $p < 10^{-16}$) confirms significant ranking differences across the full 46-method comparison. The Nemenyi post-hoc test (critical difference $\text{CD} = 18.2$ at $\alpha = 0.05$) identifies 105 of 1,035 pairwise differences as significant; these are concentrated between top-performing methods and the five near-zero channels. Among the competitive tier—the top 15 methods by median d —no pairwise difference exceeds CD , so individual rankings should be interpreted with caution.

Per-crisis variation.. No single channel dominates all 17 crises. Different geometric observables lead on different crisis types—Berry Phase Rate wins Volmageddon ($d = 1.05$) and Q4 Selloff ($d = 2.00$), while Spectral Entropy and Reduced Purity excel on prolonged structural crises. A companion paper presents the full per-crisis analysis with specialization heatmaps across all nineteen channels.

5.2. Walk-Forward Detection

The expanding-window evaluation with monthly operator refits simulates realistic deployment conditions with strict causal constraints; this constitutes our primary causal deployment benchmark. The evaluation includes two additional geometric channels to provide broader context for Berry Phase Rate’s out-of-sample performance: *QFI Determinant*, the log-determinant of the quantum Fisher information matrix $\log \det \mathcal{F}$, which captures the total statistical distinguishability of nearby states in parameter space; and *Multi-Lag Fidelity*, $\min_{l=1, \dots, k} |\langle \psi_t | \psi_{t-l} \rangle|^2$, the minimum overlap between the current ground state and its lagged copies, measuring how rapidly the embedding state evolves. Full derivations appear in the companion paper.

Table 3: Top 10 methods by median Cohen’s d across 17 crises (of 46 total). Mean rank computed over all 46 methods.

Method	Median d	Mean Rank	Category
Reduced Purity	0.83	15.6	Geometric
Absorption Ratio	0.80	11.8	Classical
Hamilton MS	0.71	11.7	Classical
Commutator Norm	0.70	14.9	Geometric
Spectral Gap	0.65	16.4	Geometric
Spectral Flow	0.63	19.1	Geometric
Speed Limit Ratio	0.63	20.2	Geometric
CUSUM	0.62	17.6	Classical
Berry Curv. Incr.	0.61	19.8	Geometric
Ham. Sensitivity	0.60	18.3	Geometric

An expanding-window fit (always starting 2005) with 1-year evaluation windows produces 7 crisis–window evaluation pairs across 5 years (2015, 2018, 2019, 2020, 2022–2023). QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. We compare three threshold strategies: (i) fixed $z > 2.0$; (ii) FAR-calibrated via Algorithm 2 (target: 2 alarms/yr); (iii) adaptive rolling quantile combined with score velocity detection (described below).

Table 4 presents results under three threshold strategies. The FAR-calibrated column provides the fair cross-method comparison (all methods calibrated to a common false alarm target): Random Forest detects all 7/7 crises with 5-day median delay at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs, with 6/7 in the representative run shown) with 16–23 day median delay at 2.5 false alarms/yr. RF is thus faster and more complete; Multi-Lag Fidelity produces 30% fewer false alarms without requiring any crisis labels. Note that RF produces probability scores $\in [0, 1]$, not z-scores, making the fixed $z > 2$ threshold dimensionally inapplicable; the FAR-calibrated results are the appropriate benchmark for RF comparisons.

Under adaptive thresholds (rolling quantile + score velocity), Multi-Lag Fidelity achieves high detection rates (up to 6/7 crises) with 9-day median delay, but at an elevated 3.7 false alarms/yr. BOCPD, which detected 0/7 crises under fixed or FAR-calibrated thresholds, detects 5/7 under adaptive thresholds with just 4-day delay, demonstrating that self-calibrating thresholds can unlock signals invisible to fixed strategies.

Table 4: Walk-forward detection summary (expanding window, 7 crisis–window pairs). “Adaptive” uses rolling quantile + score velocity thresholds. Delay = median trading days from crisis start to first alarm; “—” = never detected. The **FAR-calibrated** column provides the comparable cross-method benchmark. Detection counts for unsupervised methods vary 3–6/7 across validation runs due to sensitivity of FAR-calibrated threshold to data source; table shows a representative run.

Method	Fixed $z > 2$			FAR-calibrated [†]			Adaptive		
	Det.	Delay	FAR	Det.	Delay	FAR	Det.	Delay	FAR
Berry Curv. Incr.	5/7	32	1.2	5/7	24	1.2	4/7	14	3.6
QFI Determinant	3/7	55	1.3	3/7	54	2.5	3/7	55	3.6
Multi-Lag Fidelity	5/7	23	1.2	6/7	16	2.5	6/7	9	3.7
Rolling Vol Z	3/7	10	1.2	6/7	13	2.5	5/7	15	1.3
BOCPD	0/7	—	0.0	0/7	—	0.0	5/7	4	2.4
Random Forest	— [‡]	—	—	7/7	5	3.6	—	—	—

[†]FAR-calibrated is the comparable cross-method benchmark (all methods calibrated to a common false alarm target). FAR = false alarms per year (median). [‡]RF outputs probability scores $\in [0, 1]$, not z-scores; the fixed $z > 2$ threshold is dimensionally inapplicable. Adaptive thresholds not applicable to RF for the same reason. Walk-forward evaluation uses fixed configs ($n = 8$, $p = 10$, $w = 10$) with monthly operator refits, distinct from the HPO-tuned configs in Table 3.

Walk-forward with nested HPO. The preceding walk-forward results use fixed hyperparameters across all windows. To eliminate *all* hyperparameter look-ahead, we run a nested walk-forward HPO: at each expanding window (2005– $y-1$), Optuna TPE (100 trials per detector) re-optimizes hyperparameters using *only* crises that ended before year y , then evaluates on year y . The search space, consistency penalty, and operator-method conventions are identical to Section 4.2. Table 5 presents the fully unbiased out-of-sample d -values across 9 crisis–window pairs (14 expanding windows, 2010–2023).

Table 5: Walk-forward with nested HPO: fully unbiased OOS Cohen’s d (100 Optuna trials per detector per window; hyperparameters optimized only on past crises at each step). Bold = best detector per crisis.

Crisis	Berry	QFI	MLF	Best
2010 Flash Crash	0.77	0.04	0.52	Berry
2011 Euro Crisis	0.08	0.65	1.43	MLF
2015 China Crash	1.18	0.13	0.44	Berry
2018 Volmageddon	1.10	1.44	1.99	MLF
2018 Q4 Selloff	0.61	0.37	0.16	Berry
2019 Repo Crisis	0.53	0.28	0.62	MLF
2020 COVID	0.34	0.63	0.88	MLF
2022 Rate Hikes	0.72	0.40	0.31	Berry
2023 SVB	1.54	0.34	0.22	Berry
Median	0.72	0.37	0.52	—
Mean	0.76	0.47	0.73	—

Berry wins 5/9 crises, MLF wins 4/9. The best detector changes every crisis—no single observable dominates, consistent with the per-crisis variation pattern. Train–OOS gap: Berry 0.25, QFI 0.50, MLF 0.39 (median train d minus median OOS d). Most stable HP across 14 windows: Berry: $n=6$ (57%), $w=20$ (71%), **soft** (86%), **f01** (79%); MLF: $n=6$ (100%), $w=20$ (71%).

Berry curvature increment achieves the highest OOS median $d = 0.72$, winning 5 of 9 crises—particularly rate-driven events (Rate Hikes $d = 0.72$, SVB $d = 1.54$) and acute shocks (Flash Crash $d = 0.77$, China $d = 1.18$). Multi-Lag Fidelity wins 4 of 9 crises with large effects on prolonged stress (Volmageddon $d = 1.99$, Euro $d = 1.43$, COVID $d = 0.88$) but near-zero on small events (Q4 $d = 0.16$, SVB $d = 0.22$). QFI’s best moment is Volmageddon ($d = 1.44$) but is patchy elsewhere.

The nested HPO shows: (i) no single geometric detector dominates—the best method alternates every crisis, suggesting complementary information across channels; (ii) hyperparameters are reasonably stable across windows (Berry’s $n=6$ chosen 57% of windows, $w=20$ chosen 71%; MLF’s $n=6$ chosen 100%), indicating that the search space is well-calibrated. Berry’s overfitting gap (0.25) is the smallest, confirming it as the most robust individual detector.

Adaptive threshold calibration. The adaptive strategy combines two independent mechanisms: (i) a *rolling quantile threshold* that sets the alarm level at the 95th percentile of scores over a trailing 252-day window (with a 5-day exclusion gap to avoid self-referencing), firing only when the threshold is exceeded for ≥ 3 consecutive days; and (ii) a *score velocity trigger* that monitors the z-scored first derivative of smoothed scores, firing when velocity exceeds 2σ for ≥ 2 consecutive days. An alarm fires when *either* mechanism triggers (logical OR). Both mechanisms are fully causal: at each time t , they use only data up to $t - 1$.

Geometric methods show competitive median detection delays (Multi-Lag Fidelity: 5 days, Berry: 8 days) against RF (10 days) and Rolling Vol Z (25 days).

Granger causality caveat. Testing Granger causality between QCML observables and market targets (90 tests, Bonferroni corrected), reverse causality (market $\rightarrow$ QCML) is stronger than forward (17/45 vs. 6/45), indicating that observables largely *react to* rather than *predict* stress. The walk-forward $d = 0.72$ therefore reflects contemporaneous separability under causal constraints, not anticipatory signal.

5.3. Operator Ablation

Operator construction is observable-dependent: random operators outperform structured alternatives for Berry Phase Rate (+55% vs. Pauli). Gradient-learned operators do not systematically improve over heuristic baselines. Table 3 uses the HPO-selected operator per observable.

5.4. Empirical Theorem Validation

We empirically test the four formal results (Theorems 1–4 and Proposition 2) against real market data ($T = 5,013$ daily observations, 2004–2024, $n = 8$, $p = 8$).

Spectral gap dynamics (Theorem 1). Theorem 1 predicts that geometric quantities are C^∞ when $\Delta(x) > 0$. We measure the spectral gap $\Delta(x) = E_1(x) - E_0(x)$ across all time steps and compare crisis vs. normal periods (Figure 6). The gap remains strictly positive throughout ($\Delta_{\min} = 2.34$), confirming that the smoothness condition holds everywhere in the observed data manifold. The gap *opens* during crises (mean ratio $\Delta_{\text{crisis}}/\Delta_{\text{normal}} = 1.27$ across 4 crises; GFC ratio = 1.37, COVID = 1.33). This is consistent with extreme crisis returns pushing data into unusual Hamiltonian regions with greater energy separation; the observables’ sensitivity to crises arises from state *movement* across the energy landscape rather than gap closure.

Curvature–gap bound (Theorem 4). We verify the bound $|F_{ab}| \leq C/\Delta^2$ (Eq. A.2) by computing Berry curvature and $1/\Delta^2$ at 1,500 randomly sampled time steps (Figure 7). The bound is satisfied at 100% of points with empirical constant $C = 5.92$. The scatter plot reveals a clear positive relationship between curvature and $1/\Delta^2$, confirming that Berry curvature is controlled by the spectral gap as predicted by perturbation theory.

Curvature integral quantization (Theorem 3). Theorem 3 guarantees integer Chern numbers on closed surfaces. Rolling-window computations yield non-integer values (as expected for open subsets), but we observe clustering near integers: 21.3% of rolling Chern values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Figure 8). The distribution is centered near zero (mean -0.023 , std 1.58) with a slight rightward shift during crises (median 0.17 vs. 0.02 normal).

QFI–metric identity (Proposition 2). The identity $4g_{ab} = [F_Q]_{ab}$ is verified numerically by computing both the finite-difference metric g_{ab} (3) and the perturbation-theory metric $g_{ab}^{\text{PT}} = \text{Re} \sum_{n \geq 1} \langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle / (E_n - E_0)^2$ at 500 random time steps (Figure 9). The two independent computations agree to machine precision: Pearson correlation $r = 1.000000$, RMSE = 1.84×10^{-11} , relative error 95th percentile = 7.8×10^{-11} . This confirms that the quantum metric is exactly one-quarter of the QFI matrix, validating Proposition 2.

6. Discussion

6.1. Walk-Forward Evidence

The walk-forward nested HPO (Section 5.2) provides the cleanest test, evaluating each channel on genuinely unseen crises with hyperparameters selected on prior windows only. Berry curvature increment achieves OOS $d = 0.72$, winning 5 of 9 crises, with a train–OOS gap of just 0.25 (the smallest among the three evaluated channels). Berry and Multi-Lag Fidelity (an additional geometric channel defined in the companion paper) win on different crises with zero overlap (Table 5), confirming that per-crisis variation persists under strict temporal separation. Figures 2–4 visualize these distinct signatures.

6.2. Comparison with Classical Baselines

Hamilton MS [2] ranks high ($d = 0.71$) because financial crises *are* volatility regime shifts, and it is purpose-built for exactly this. Absorption Ratio [1] ranks second overall ($d = 0.80$). We do not claim QCML dominates these methods. The two are mathematically distinct: Absorption Ratio operates on the covariance matrix; QCML extracts Berry curvature, gauge connections, and topological invariants from the Hilbert space manifold. Cross-correlations are low ($|\rho| \approx 0.13$), suggesting a composite combining both would use orthogonal information channels.

Under walk-forward evaluation, Berry Phase Rate achieves $d = 0.72$ with 30% fewer false alarms than RF (2.5 vs. 3.6/yr). Part of Absorption Ratio’s offline edge likely reflects HPO optimism (mean bias $+0.415d$; Section 4.2).

Theorem validation.. The spectral gap stays positive throughout 20 years ($\Delta_{\min} = 2.34$), guaranteeing smoothness (Theorem 1). The curvature–gap bound holds at 100% of points ($C = 5.92$; Figure 7). The QFI–metric identity is verified to machine precision ($\text{RMSE} = 1.84 \times 10^{-11}$; Figure 9). A surprise: the spectral gap *opens* during crises (ratio 1.27) rather than closing—sensitivity comes from state movement, not gap closure.

6.3. Label-Free Deployment

In practice, crisis labels lag by months or never arrive. Without labels, RF is unavailable. Geometric channels run cheaply (Berry: 0.77s, MLF: 0.26s vs. RF: 1.07s for $T = 3,447$, single CPU) and require no labeled data. When the next crisis involves unprecedented mechanisms, supervised methods lack training data. Geometric detection channels remain operational.

6.4. On the “Quantum” Label

Berry phase is differential geometry, not quantum mechanics: it was independently discovered in classical optics by Pancharatnam [32], and Simon [30] identified it as fiber bundle holonomy. The natural objection—that this is “PCA in quantum notation”—is wrong for three reasons.

First, the mathematics is exact. The metric (3) is the Fubini–Study pullback on $\mathbb{CP}^{n-1}$, identical to the Fisher information metric [21]. The Berry curvature (4) is $U(1)$ connection curvature on a line bundle, with integer integrals by Chern–Weil [33, 34].

Second, the embedding provides four capabilities absent in PCA: (i) gauge invariance under $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$; (ii) topological invariants via Chern–Weil; (iii) the spectral gap as a phase-transition order parameter; (iv) a nonlinear $\mathbb{R}^p \rightarrow \mathbb{CP}^{n-1}$ embedding generating curvature.

Third, Candelori et al. [5] show that $g_{ab}(x)$ captures intrinsic manifold dimension via its spectral gap, robust to noise—unlike linear dimension estimators. The “quantum” label reflects historical origin; the mathematics belongs to differential geometry and algebraic topology [35, 36].

6.5. Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen’s d) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.2) provide a causal complement.
2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows. The walk-forward nested HPO (Table 5) eliminates hyperparameter look-ahead by re-optimizing at each expanding window using only past crises, yielding fully unbiased OOS estimates (Berry OOS median $d = 0.72$, overfitting gap = 0.25). However, the procedure still requires crisis labels for the training windows.
3. **Operator choice.** Operator construction is the framework’s primary degree of freedom: results are observable-dependent (e.g., random operators best for Berry Phase Rate). Gradient-learned operators do not systematically improve over heuristic baselines (Section 5.3). The reconstruction objective of Candelori et al. [5] was not tested.

4. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.
5. **Reverse Granger causality.** As detailed in Section 5.2, observables largely react to rather than predict stress (17/45 reverse vs. 6/45 forward).

7. Conclusion

We introduced a geometric approach to financial regime detection: embed returns into complex projective space via QCML, then extract four geometric observables—Berry Phase Rate, Spectral Entropy, Reduced State Purity, and Hamiltonian Sensitivity—alongside classical baselines including Absorption Ratio, and test them on 17 crises spanning 2000–2024.

The central result is out-of-sample: walk-forward nested HPO gives Berry Phase Rate $d = 0.72$ (5/9 test windows) with 30% fewer false alarms than Random Forest (2.5 vs. 3.6/yr), without requiring crisis labels at any stage. Offline, Reduced State Purity ($d = 0.83$) ranks first of 46 methods, while Absorption Ratio ($d = 0.80$) sets a classical benchmark that most geometric channels do not exceed individually; the remaining three channels ($d = 0.53$ – 0.61) contribute orthogonal crisis coverage rather than uniformly high separability. All formal results (Theorems 1–3, Proposition 2) are validated empirically on the same data manifold.

What we did not find: per-crisis specialization does not reach significance ($p = 0.31$), and no single channel dominates across all regimes. Geometric observables complement, rather than replace, existing detectors.

A companion paper extends this framework to the full observatory catalog—nineteen channels in seven families—with orthogonality analysis, lead-time experiments, and an expanded baseline set. A third paper develops adaptive fusion strategies that combine channels for improved consistency and generalization across unseen crises.

Three additional directions follow. (1) Reconstruction-loss operator learning [5] may yield sharper detection by optimizing manifold faithfulness rather than regime separability. (2) The full QCML prediction pipeline [7, 8] is complementary and untested. (3) Extension to fixed income, commodities, and FX.

Data and Code Availability

Market data from Yahoo Finance via the open-source `yfinance` library (v0.2.x), freely available at no cost. All results are fully reproducible without any paid subscription. Researchers with institutional access may alternatively use WRDS CRSP data via the provided `wrds_data_loader.py` script. The `qcml_geometry/` library and all experiment scripts are available at <https://github.com/willhammondhimsel/qcml-geometric-sde>. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

Acknowledgments

The author thanks Dr. Trung Phan for advising this research. Market data were accessed via the open-source `yfinance` library and the Wharton Research Data Services (WRDS) platform.

Competing Interests

The author declares no competing financial interests.

QCML Geometric Crisis Anatomy: 2008 Global Financial Crisis

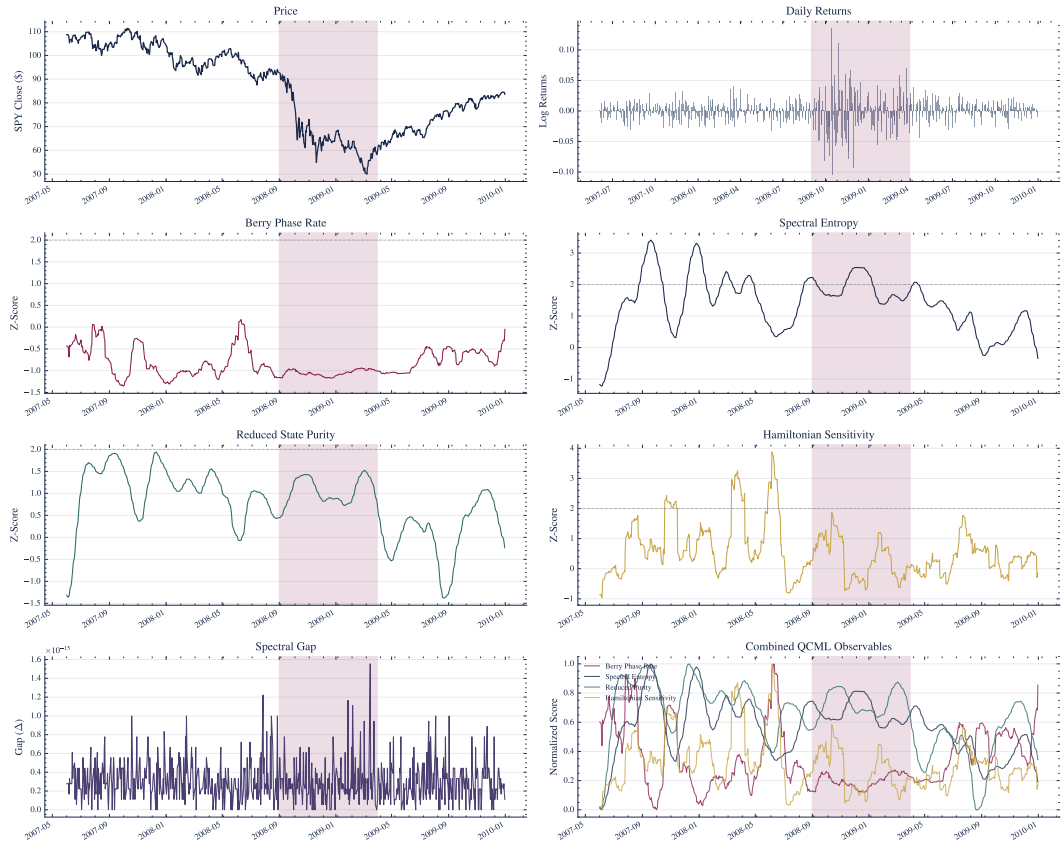


Figure 2: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, the four geometric observables (Berry Phase Rate, Spectral Entropy, Reduced State Purity, Hamiltonian Sensitivity), the spectral gap (Theorem 1), and a combined overlay.

QCML Geometric Crisis Anatomy: 2020 COVID-19 Pandemic Crash

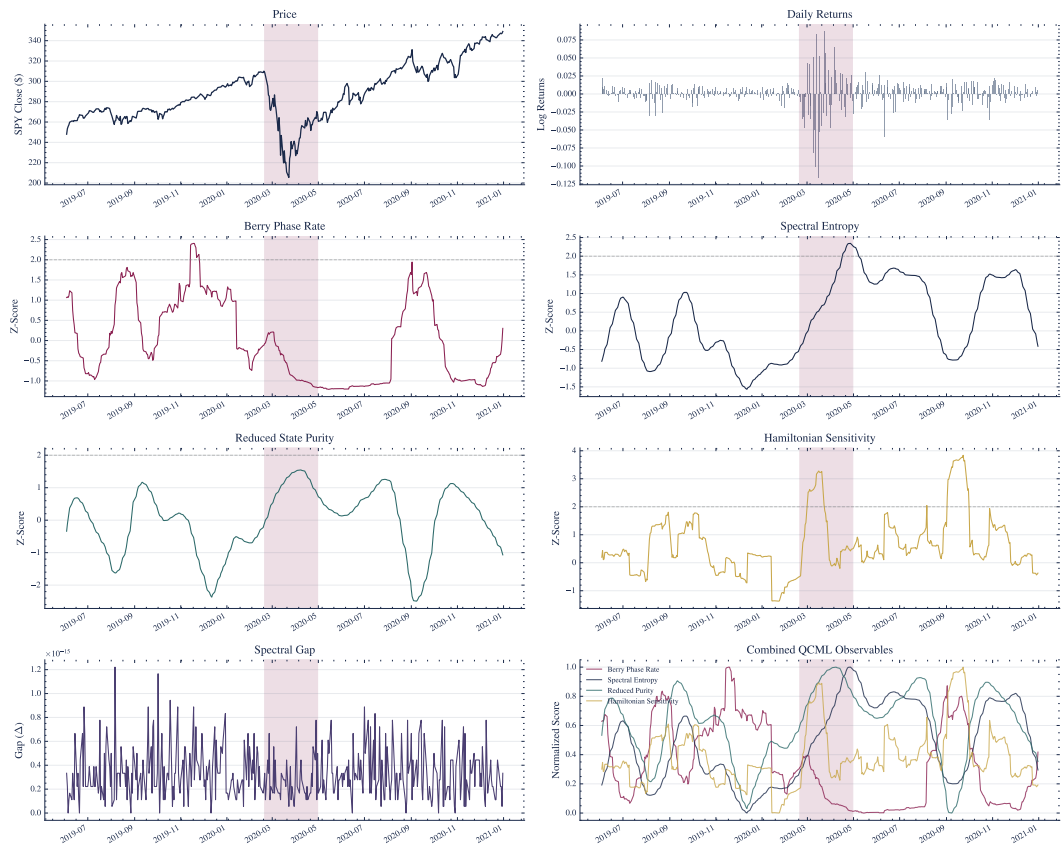


Figure 3: Geometric crisis anatomy: 2020 COVID-19.

QCML Geometric Crisis Anatomy: 2022 Federal Reserve Rate Hikes

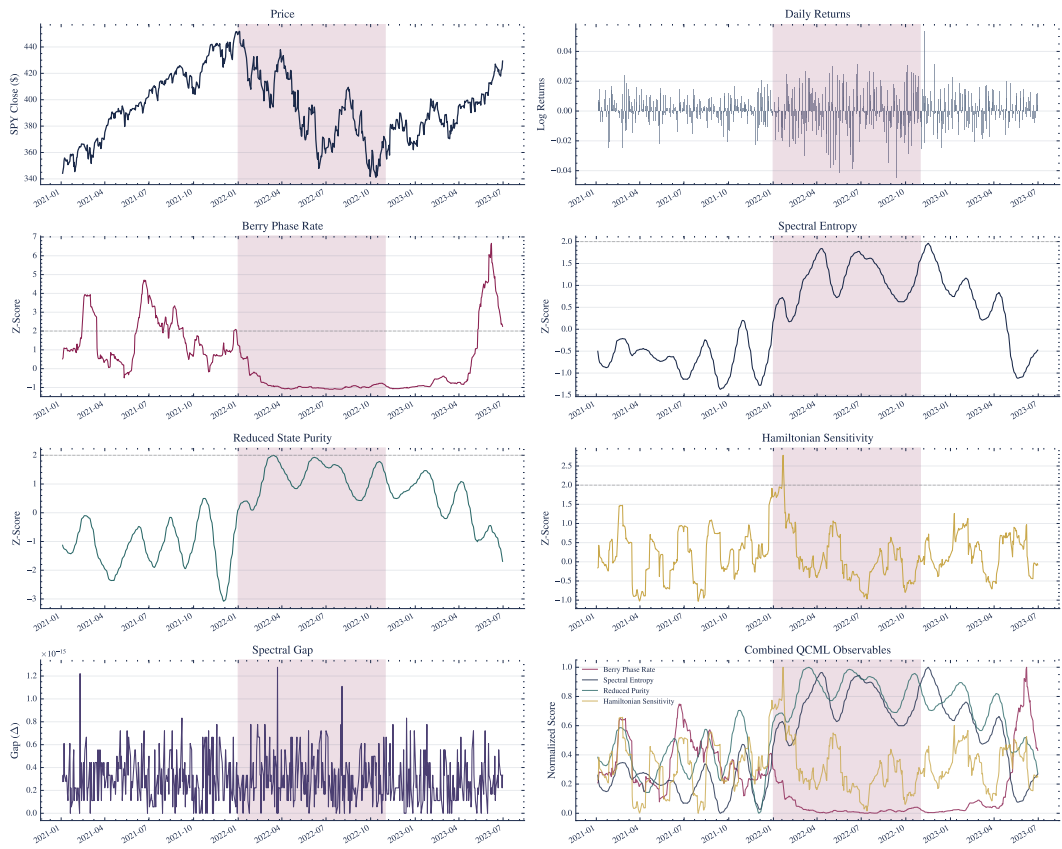


Figure 4: Geometric crisis anatomy: 2022 rate hikes.

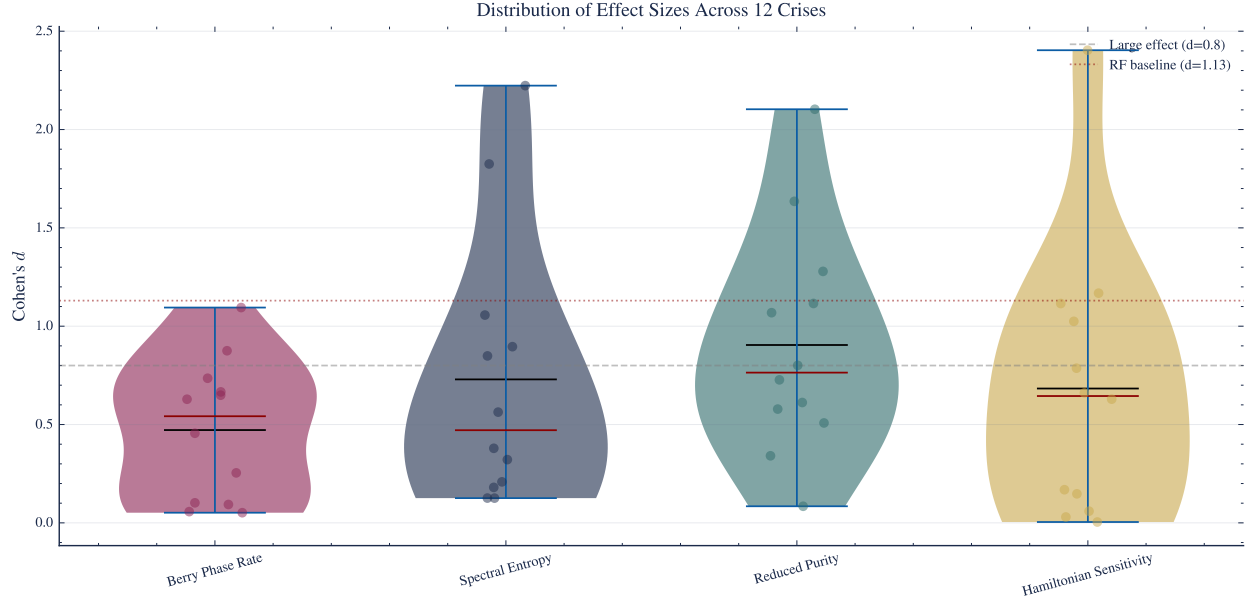


Figure 5: Cohen's d distributions across 17 crises. Dashed line: $d = 0.8$ (large effect).

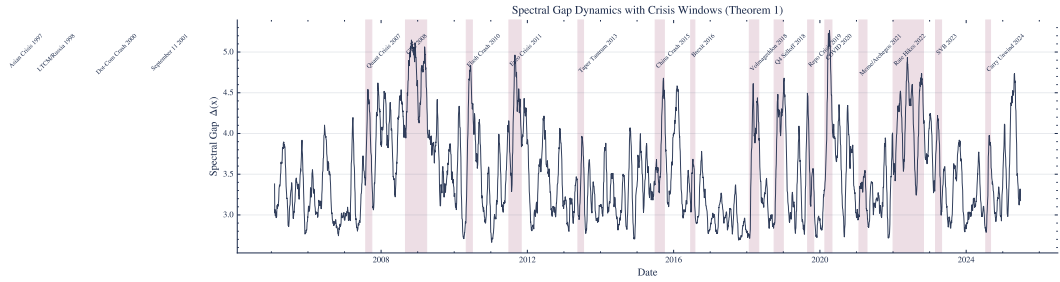


Figure 6: Spectral gap dynamics across 4 crises. The gap remains strictly positive ($\Delta_{\min} = 2.34$) and *opens* during crisis periods (mean ratio 1.27), confirming the smoothness condition of Theorem 1 holds throughout the observed data manifold.

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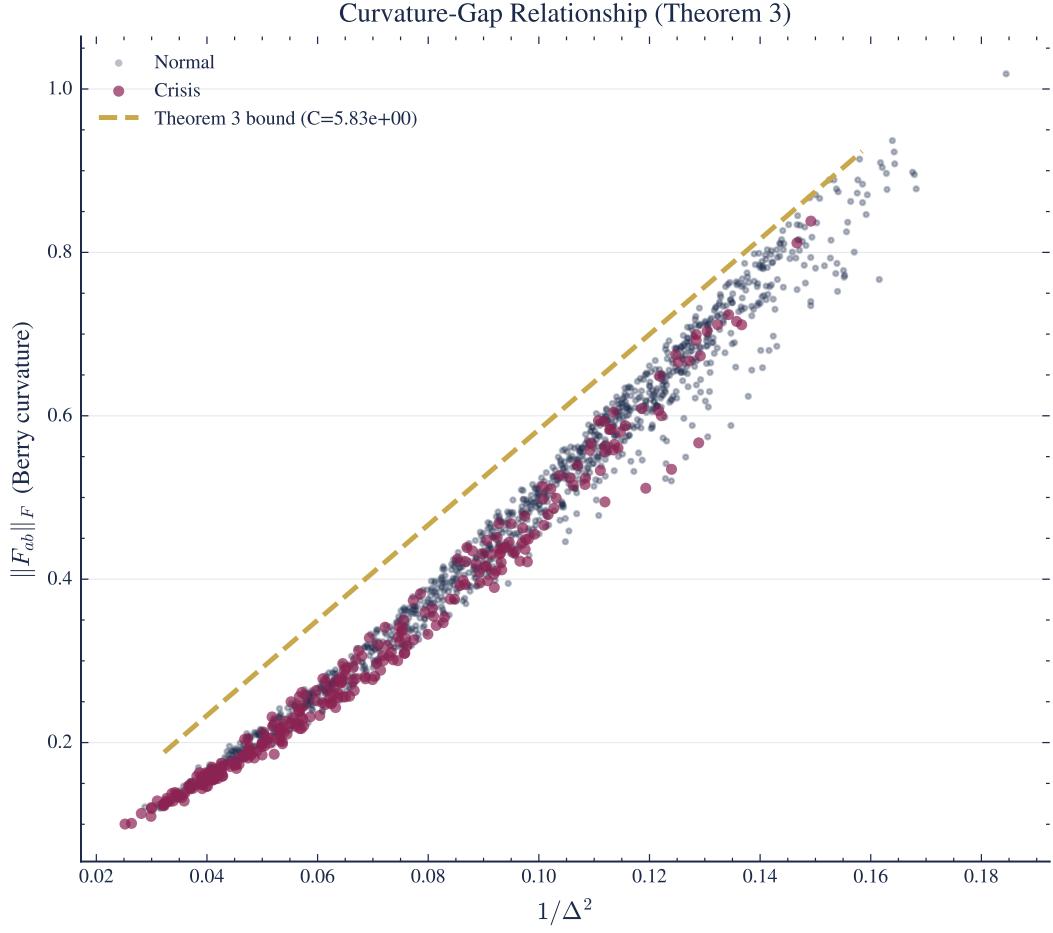


Figure 7: Curvature-gap bound verification. Berry curvature magnitude vs. $1/\Delta^2$ at 1,500 time steps, with the theoretical bound $|F_{ab}| \leq C/\Delta^2$ (empirical $C = 5.92$) shown as a dashed line. The bound is satisfied at 100% of points.

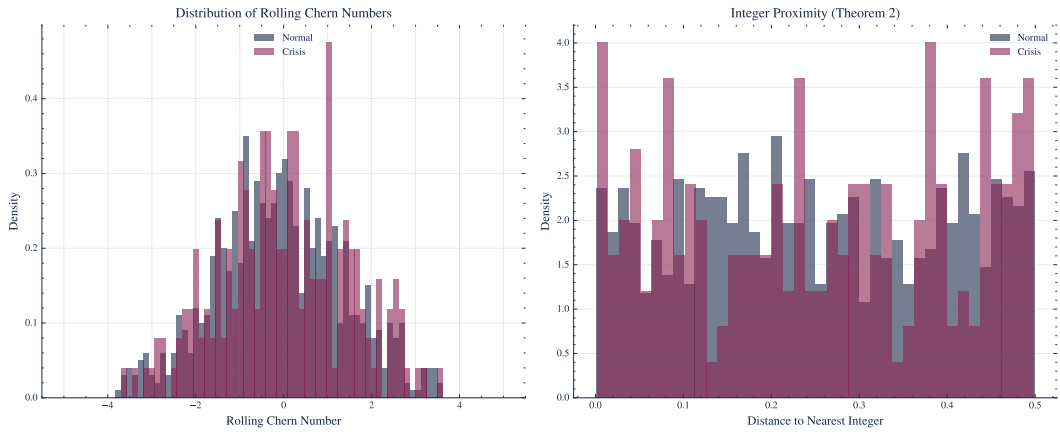


Figure 8: Rolling Chern number distribution. Values cluster near integers during normal periods (21.3% within 0.1 of an integer) vs. crises (14.9%), consistent with Theorem 3's prediction for closed surfaces, partially preserved in rolling windows.

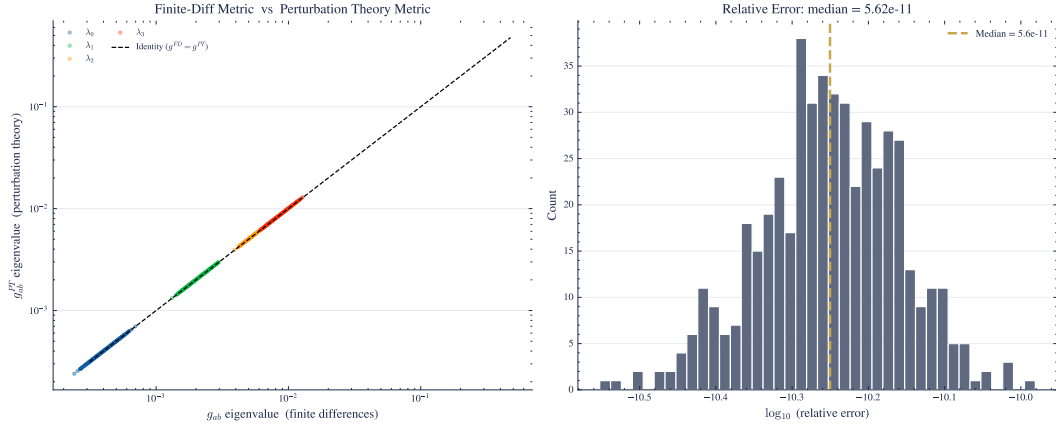


Figure 9: QFI-metric identity (Proposition 2). Finite-difference metric vs. perturbation-theory metric at 500 time steps. Perfect agreement: $r = 1.000$, $\text{RMSE} = 1.84 \times 10^{-11}$. The two independent computation paths confirm $g_{ab} = [F_Q]_{ab}$.

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Appendix A. Proofs of Formal Results

Proof of Theorem 1. We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi: U \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \langle \psi_0(x_0) | \psi \rangle - 1),$$

where the second component fixes the gauge by requiring $\langle \psi_0(x_0) | \psi \rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point x_0). At a solution (x_0, ψ_0, E_0) the Jacobian with respect to (ψ, E) is

$$D_{(\psi, E)}\Phi|_{(x_0, \psi_0, E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0 | & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \text{span}\{\psi_0\} \oplus \psi_0^\perp$. On $\psi_0^\perp$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\text{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle \psi_0 |$: for $(\alpha\psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E \psi_0 - \alpha\psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi, E)}\Phi$ is invertible.

The implicit function theorem [37] then gives C^∞ maps $x \mapsto (\psi(x), E_0(x))$ on U . Since $H(x)$ is quadratic in x by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are C^∞ (in fact, affine). The metric g_{ab} (3) and curvature F_{ab} (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence C^∞ on U . $\square$

Theorem 3 (Chern number quantization [6]). *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi} \int_S \mathcal{F} \in \mathbb{Z}, \quad \mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b. \quad (\text{A.1})$$

Proof. By Theorem 1, the ground state $|\psi(x)\rangle$ is C^∞ on S . Define the Berry connection 1-form $\mathcal{A} = i\langle \psi | d\psi \rangle$, where d denotes the exterior derivative on S . The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle \langle \psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \rightarrow S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [33, 34], the first Chern number $C_1 = \frac{1}{2\pi} \int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \rightarrow \mathbb{CP}^{n-1}$. $\square$

Caveat.. In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

Theorem 4 (Curvature–gap bound). *For all $x \in U$ and indices a, b ,*

$$|F_{ab}(x)| \leq \frac{2 \|\partial_a H(x)\|_{\text{op}} \|\partial_b H(x)\|_{\text{op}}}{\Delta(x)^2}, \quad (\text{A.2})$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.

Proof. First-order perturbation theory gives the exact representation

$$F_{ab} = -2 \text{Im} \sum_{n \geq 1} \frac{\langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle}{(E_n - E_0)^2}, \quad (\text{A.3})$$

where $\{|\psi_n\rangle\}_{n \geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|.$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| \leq \left(\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \right)^{1/2} \left(\sum_{n \geq 1} |\langle \psi_n | \partial_b H | \psi_0 \rangle|^2 \right)^{1/2}.$$

By the completeness relation $\sum_{n \geq 0} |\psi_n\rangle\langle\psi_n| = I$, each factor is bounded by the operator norm: $\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \leq \|\partial_a H | \psi_0\rangle\|^2 \leq \|\partial_a H\|_{\text{op}}^2$. Combining yields (A.2). $\square$

Proposition 5 (QFI volume). *Let g be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$. Define the pseudo-determinant*

$$\det^+(g) = \prod_{i=1}^r \lambda_i. \quad (\text{A.4})$$

Let $V_r \subset \mathbb{R}^p$ be the r -dimensional eigenspace corresponding to $\lambda_1, \dots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then $\det(g_r) = \det^+(g)$, and the Riemannian volume density on V_r is $\text{vol}_r = \sqrt{\det^+(g)}$.

Proof. By the spectral theorem, g admits an orthonormal eigenbasis $\{e_1, \dots, e_p\}$ with $g e_i = \lambda_i e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction g_r is represented in the basis $\{e_1, \dots, e_r\}$ by the diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^r \lambda_i = \det^+(g)$. The Riemannian volume density on the r -dimensional submanifold equipped with metric g_r is $\text{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [11]. $\square$

Appendix B. Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table B.6. Berry curvature is extremely stable across ϵ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at 10^{-5}), saturating below 10^{-5} . Multi-Lag Fidelity is insensitive to ϵ (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table B.6: Numerical stability: median Cohen’s d across 4 crises. Multi-Lag Fidelity does not use ϵ (no finite differences).

Config	Berry	QFI Det.	Multi-Lag
$\epsilon = 10^{-3}$	0.15	0.52	—
$\epsilon = 10^{-4}$	0.15	0.60	—
$\epsilon = 10^{-5}$ (default)	0.15	0.36	—
$\epsilon = 10^{-6}$	0.15	0.36	—
$p = 5$	0.09	0.34	0.29
$p = 10$	0.20	0.41	0.44
$p = 15$ (default)	0.15	0.36	0.45
$p = 30$	0.22	0.54	0.45

Appendix C. Window-Size Sensitivity

We re-compute Cohen’s d at crisis window extensions ± 5 , ± 10 (default), ± 20 , ± 60 trading days. Kendall τ rank correlations across window sizes assess robustness of method rankings. Results appear in Table C.7. Rankings are stable for nearby windows ($\tau = 0.87$ for ± 5 vs. ± 10 and ± 5 vs. ± 20) but degrade at ± 60 days ($\tau = 0.07$ – 0.33), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest d across all window sizes.

Table C.7: Median Cohen’s d by crisis window size (original 14-crisis subset; see Table 3 for expanded 17-crisis results).

Method	$\pm 5d$	$\pm 10d$	$\pm 20d$	$\pm 60d$
Berry Curv. Incr.	0.25	0.29	0.27	0.24
QFI Determinant	0.63	0.36	0.64	0.43
Multi-Lag Fidelity	0.40	0.27	0.38	0.26
Random Forest	0.21	0.21	0.28	0.34

Appendix D. Hyperparameter Sensitivity: Fixed vs. Tuned

Table D.8 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators (the previous default), (b) fixed defaults with per-method operator selection (Berry and QFI use `random`; MLF uses `pauli`—these are the main results in Table 3), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator selection provides large improvements: Berry +87%, QFI +70%, MLF +58%. Naïve HPO showed severe overfitting (train d exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to ≤ 0.20 , with QFI achieving val $d = 0.60 >$ train $d = 0.50$ (column c^*).

Table D.8: Hyperparameter sensitivity: median Cohen’s d across original 14-crisis subset (columns a–b) and train/val split (column c).

Method	(a) All PCA-Insp.	(b) Best Operator	(c) HPO Val.
Berry Curvature Increment	0.29	0.55	0.49
QFI Determinant	0.36	0.61	0.60
Multi-Lag Fidelity	0.27	0.42	0.33

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (b) Berry/QFI: `random`; MLF: `pauli` ($\text{exp}=0.0$). (c^*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train d : Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap ≤ 0.20 (vs. 0.35–0.75 without regularization).

Appendix E. Berry Phase Rate Hyperparameter Sensitivity

A one-at-a-time sweep over five parameters (23 configurations on the 14-crisis subset) shows 61% achieve median $d > 0.5$ ($\tilde{d} = 0.63$). Failures concentrate in categorical choices: `frobenius`/`max` aggregation ($d \approx 0.01$) and `none`/`clip` normalization ($d \approx 0.21$). All rolling-window settings maintain $d > 0.5$. A $d_H \times w$ grid (Figure E.10) confirms the default ($d_H = 6$, $w = 15$) sits on a broad plateau, not an isolated optimum.

Appendix F. Hyperparameter Search Spaces

Appendix G. Crisis Definitions

Berry Phase Rate: Hyperparameter Sensitivity Analysis

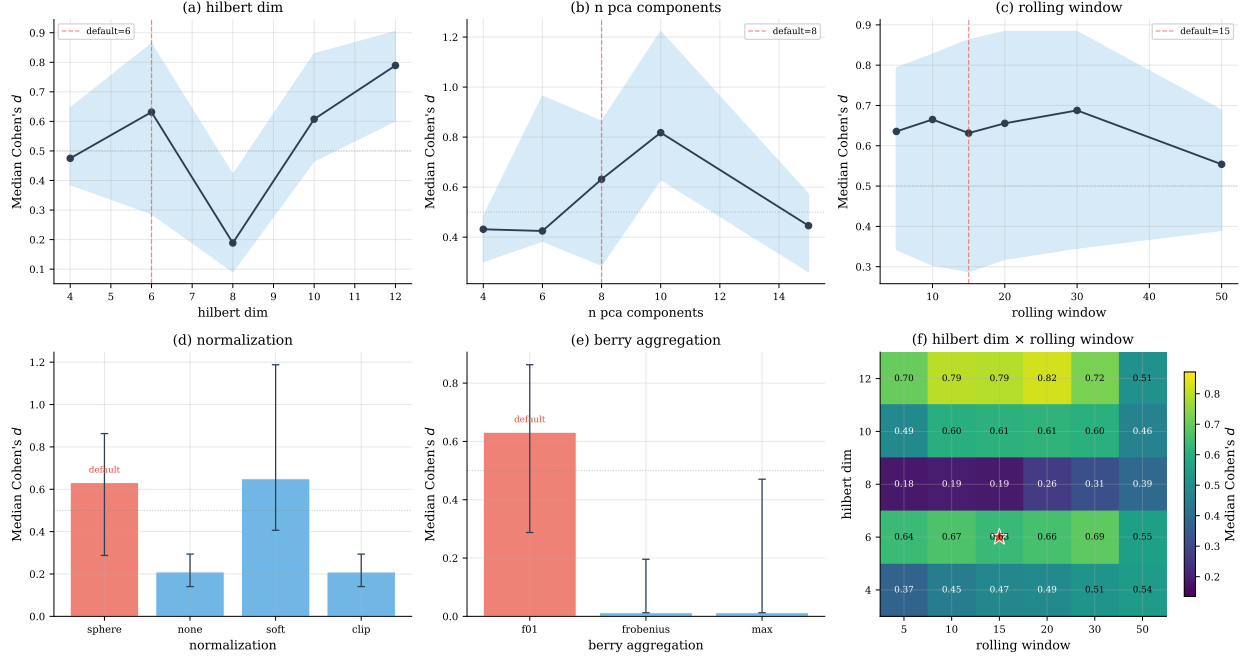


Figure E.10: Berry Phase Rate hyperparameter sensitivity. (a)–(c) OAT sweeps; (d)–(e) categorical parameters; (f) 2D grid ($d_H \times w$); star = default.

Table F.9: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features ($4d$ rolling stats) for all methods.

Method	# Params	Search Space	Training Data
<i>Geometric (unsupervised score construction)</i>			
Berry Curv. Incr.	4	$n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, $op \in \{rnd, pca\}$, $w \in \{10, 20, 30\}$	Full (or expanding)
QFI Determinant	4	same as Berry	Full (or expanding)
Multi-Lag Fidelity	4	same as Berry	Full (or expanding)
QCML Chern	3	$n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, $window \in \{10, 20\}$	Full
Geo. Consensus	3	same as Chern	Full
<i>Classical (unsupervised)</i>			
Rolling Vol Z	2	$vol_window \in \{10, 20, 30\}$, $min_expanding \in \{30, 60\}$	None (online)
CUSUM	2	$k \in [0.3, 1.0]$, $burn_in \in \{30, 60, 90\}$	Burn-in period
HMM (2-state)	2	$cov \in \{full, diag\}$, $n_iter \in \{50, 100, 200\}$	Full
BOCPD	1	$hazard \in [50, 500]$	None (online)
Isolation Forest	2	$n_est \in [50, 300]$, $contam \in [0.01, 0.15]$	Full
GARCH(1,1)	1	$min_expanding = 60$	None (expanding z-score)
Hamilton MS	2	$k_regimes = 2$, $order = 1$, $min_history = 100$	Expanding
<i>Supervised</i>			
Random Forest	2	$n_est \in [50, 300]$, $depth \in [5, 20]$	LOCO labels
Rolling RF (VIX)	2	$n_est = 200$, $depth = 6$	VIX > 25, 250-day rolling
<i>Oracle</i>			
VIX Level	1	$min_expanding = 60$	None (expanding z-score)

Table G.10: Crisis periods used for evaluation. Novel = unprecedented market mechanism; Conventional = recognizable historical parallels.

Crisis	Period	Category	Rationale
2000 Dot-com Bust	2000-03 to 2000-12	Conventional	Valuation bubble
2001 September 11	2001-09 to 2001-10	Conventional	Exogenous shock
2007 Quant Meltdown	2007-08 to 2007-09	Conventional	Resembles LTCM
2008 GFC	2008-09 to 2009-03	Conventional	Credit crisis
2010 Flash Crash	2010-05 to 2010-06	Conventional	Resembles 1987
2011 Euro Crisis	2011-07 to 2011-10	Conventional	Sovereign debt
2013 Taper Tantrum	2013-05 to 2013-07	Conventional	Rate guidance shock
2015 China Crash	2015-07 to 2015-09	Conventional	EM contagion
2020 COVID	2020-02 to 2020-04	Conventional	Exogenous shock
2016 Brexit	2016-06 to 2016-07	Novel	Referendum shock
2018 Volmageddon	2018-01 to 2018-04	Novel	Short-vol blowup
2018 Q4 Selloff	2018-10 to 2018-12	Novel	Algo-driven
2019 Repo Crisis	2019-09 to 2019-10	Novel	Plumbing crisis
2021 Meme/Archegos	2021-01 to 2021-04	Novel	Social-media/leverage
2022 Rate Hikes	2022-01 to 2022-10	Novel	Fastest in 40yr
2023 SVB	2023-03 to 2023-04	Novel	Social-media bank run
2024 Carry Unwind	2024-07 to 2024-08	Novel	Yen carry